Angle bisectors and perpendicular bi[redacted]ors

1

a A point is on the perpendicular bisector of a line segment if and only if it is equidistant from the end points of the line segment.

b A point is on the bisector of an angle if and only if it is equidistant from the two sides of the angle.

2

a

i $x = 9$

ii The two line segments are congruent by the "angle bisector theorem".

b

i $x = 12$

ii The vertical line segment bisects the horizontal line segment by the "converse of perpendicular bisector theorem".

c

i $x = 7$

ii The peak of the triangle is equidistant from the two other vertices by the "perpendicular bisector theorem".

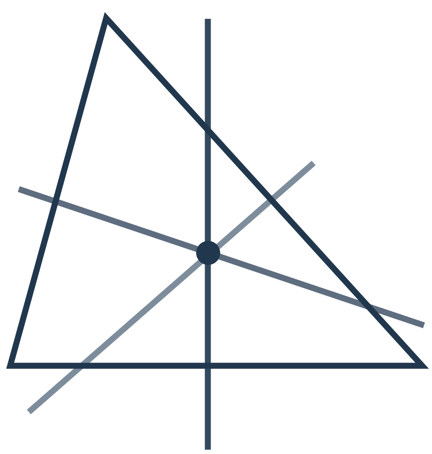
d

i $x = 23$

ii The vertical line segment is an angle bisector by the "converse of angle bisector theorem".

3

a



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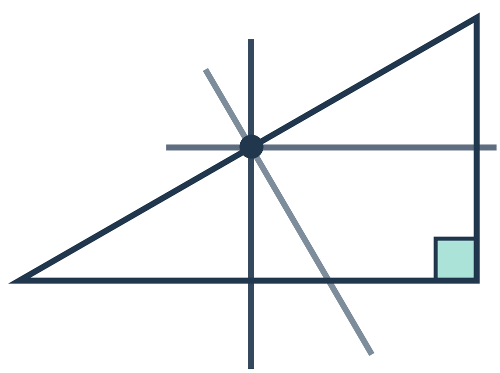

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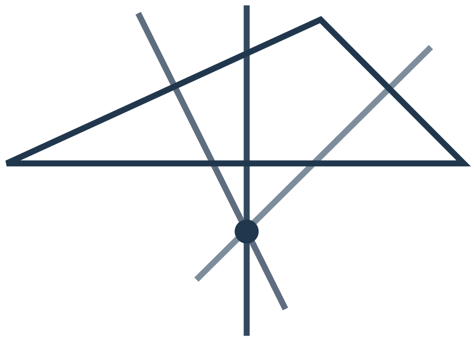
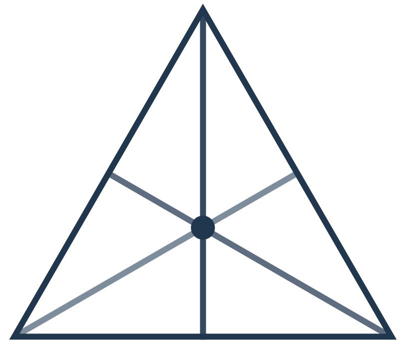
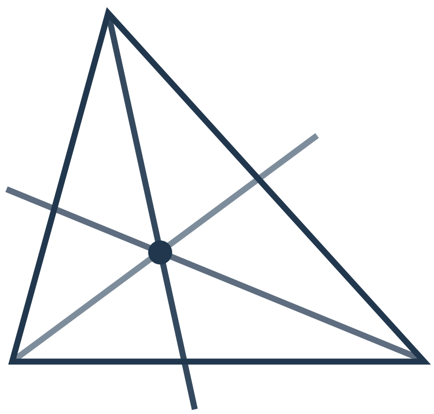


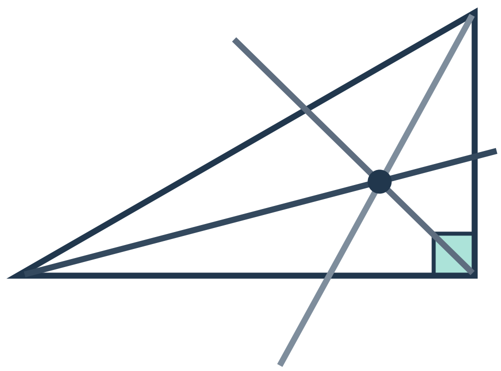


4

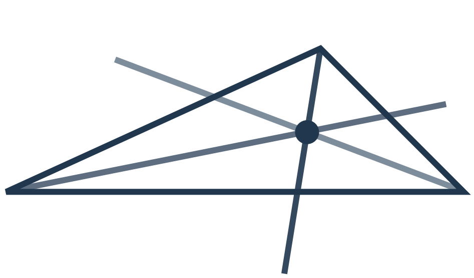
a



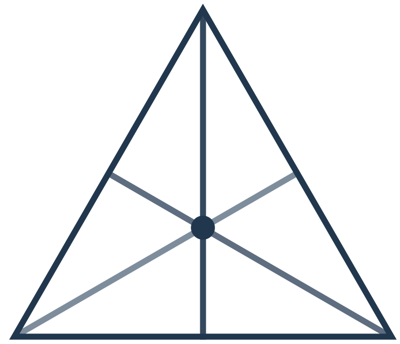
b



c



d



5

a

i Angle bisector theorem and AAS (or HL)

ii $x = 3$

b

i Converse of perpendicular bisector theorem and SSS (or HL)

ii $x = 53$

c

i Perpendicular bisector theorem and SSS (or HL)

ii $x = 11$

d

i Converse of angle bisector theorem and SSS (or AAS, or HL)

ii $x = 7$

6

a Not valid. The line segment not being bisected contradicts the converse of perpendicular bisector theorem.

b Not valid. The angle not being bisected contradicts the converse of angle bisector theorem.

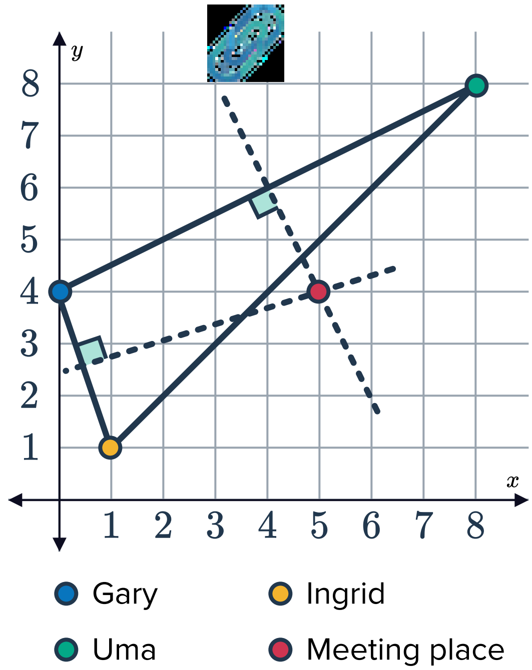
c Valid

d Not valid. The angles not being equal imply that they are not corresponding angles in congruent triangles, which implies that the bottom line segment is not bisected. This contradicts the converse of perpendicular bisector theorem.

7

a To find the location that is equidistant from three distinct points, we can find the circumcenter of the triangle formed by those three points.
So Gary, Uma and Ingrid can find their meeting place by considering the triangle joining their homes together and finding the intersection of the perpendicular bisectors of the sides of that triangle.

b



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To prove: $PA = PB$

	Statements	Reasons
1.	$AM = BM$	Given
2.	$\triangle AMP$ is a right triangle	$\overline{PM} \perp \overline{AB}$
3.	$\triangle BMP$ is a right triangle	$\overline{PM} \perp \overline{AB}$
4.	$PA^2 = AM^2 + PM^2$	Pythagorean theorem in $\triangle AMP$
5.	$PB^2 = BM^2 + PM^2$	Pythagorean theorem in $\triangle BMP$
6.	$PB^2 = AM^2 + PM^2$	Substitution property of equality
7.	$PA^2 = PB^2$	Transitive property of equality
8.	$PA = PB$	Square root of both sides of the equation

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To prove  $= BM$

	Statements	Reasons
1.	$PA = PB$	Given
2.	$\triangle AMP$ is a right triangle	$\overline{PM} \perp \overline{AB}$
3.	$\triangle BMP$ is a right triangle	$\overline{PM} \perp \overline{AB}$
4.	$PA^2 = AM^2 + PM^2$	Pythagorean theorem in $\triangle AMP$
5.	$PB^2 = BM^2 + PM^2$	Pythagorean theorem in $\triangle BMP$
6.	$PA^2 = BM^2 + PM^2$	Substitution property of equality
7.	$AM^2 + PM^2 = BM^2 + PM^2$	Transitive property of equality
8.	$PA^2 = PB^2$	Converse of addition property of equality
9.	$PA = PB$	Square root of both sides of the equation

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To prove: $SP = SR$

	Statements	Reasons
1.	$m\angle SPQ = 90^\circ$	$\overline{SP} \perp \overline{PQ}$
2.	$m\angle SRQ = 90^\circ$	$\overline{SR} \perp \overline{PQ}$
3.	$m\angle SPQ = m\angle SRQ$	Transitive property of equality
4.	$m\angle PQS = m\angle RQS$	$\overline{QS}$ bisects $\angle PQR$
5.	$\overline{QS}$ is common in $\triangle PQS$ and $\triangle RQS$	Given
6.	$\triangle PQS \cong \triangle RQS$	Congruence test AAS
7.	$SP = SR$	Corresponding sides in congruent triangles have the same length

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To prove: $m\angle PQS = m\angle RQS$

	Statements	Reasons
1.	$m\angle SPQ = 90^\circ$	$\overline{SP} \perp \overline{PQ}$
2.	$m\angle SRQ = 90^\circ$	$\overline{SR} \perp \overline{PQ}$
3.	$SP = SR$	Given
4.	$\overline{QS}$ is common in $\triangle PQS$ and $\triangle RQS$	Given
5.	$\triangle PQS \cong \triangle RQS$	Congruence test HL
6.	$m\angle PQS = m\angle RQS$	Corresponding angles in congruent triangles have the same measure

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To prove: X lies on the perpendicular bisector of BC

	Statements	Reasons
1.	X lies on the perpendicular bisector of AB	Given
2.	$AX = BX$	Perpendicular bisectors theorem
3.	X lies on the perpendicular bisector of AC	Given
4.	$AX = CX$	Perpendicular bisectors theorem
5.	$BX = CX$	Transitive property of equality
6.	X lies on the perpendicular bisector of BC	Converse of perpendicular bisectors theorem

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To prove: X lies on the angle bisector of $\angle ACB$

	Statements	Reasons
1.	X lies on the angle bisector of $\angle ABC$	Given
2.	$DX = FX$	Angle bisectors theorem
3.	X lies on the angle bisector of $\angle BAC$	Given
4.	$EX = FX$	Angle bisectors theorem
5.	$DX = EX$	Transitive property of equality